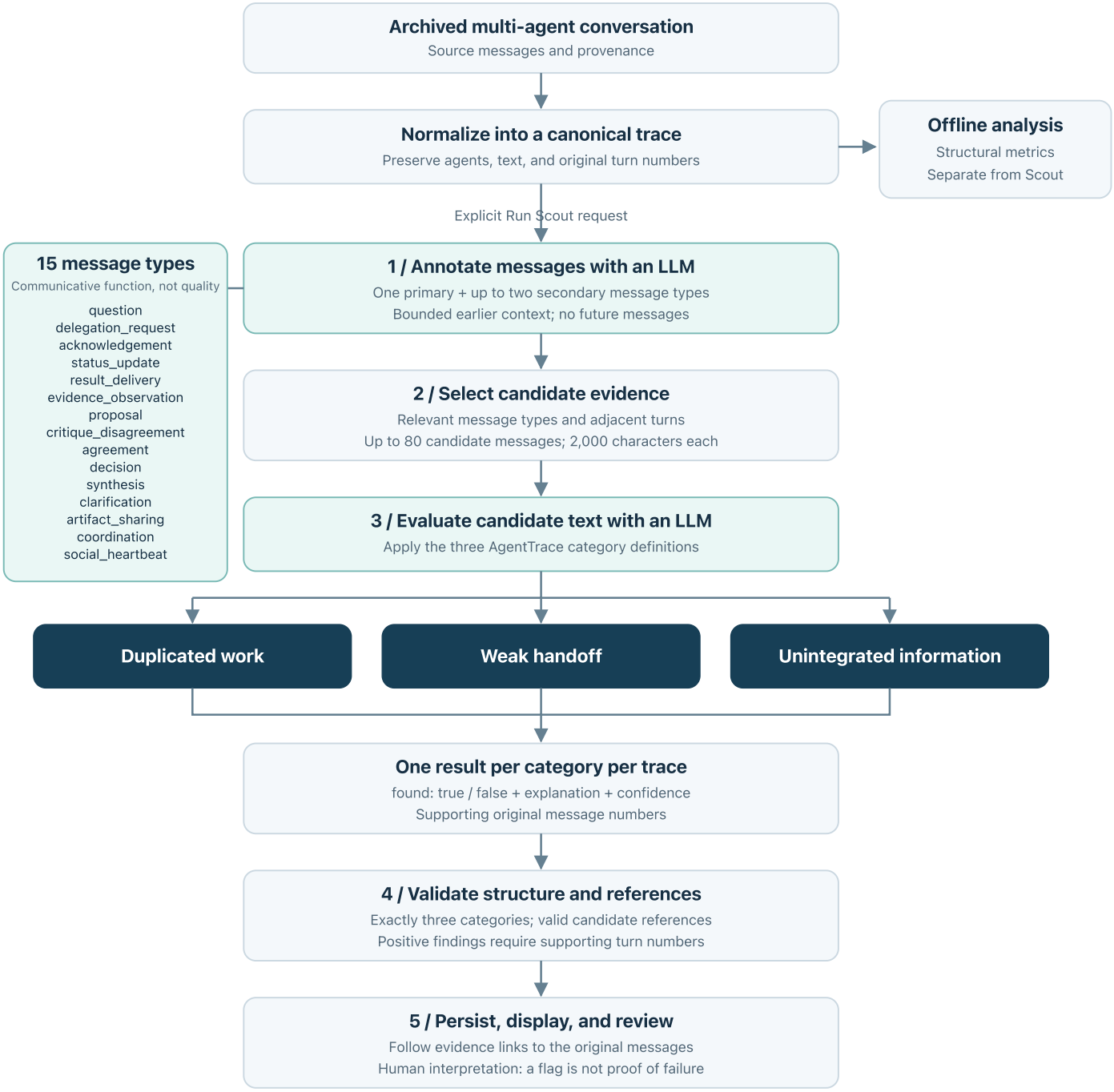
AGENTTRACE / METHOD EXPLAINER

# From conversation traces to collaboration judgments

Scout supplies the model-based analysis machinery. AgentTrace defines the three categories, selects evidence, validates references, and displays the findings.

## 1. Analysis process

Green stages use an LLM. Structural metrics are a separate offline analysis, not inputs that mechanically determine the three judgments.




Sources: method documentation · scanner implementation · separate structural metric definitions.

## 2. Scout message types

AgentTrace's Scout annotation prompt assigns **one primary type and up to two secondary types** to each target message. These labels describe **observable communicative function, not quality or success**. Annotations also include inferred replies, addressees, confidence, ambiguity, rationale, and verbatim evidence spans.

Definition source: message-type definitions and annotation rules; output fields: MessageAnnotation schema.

| Message type | Definition used in the prompt |
|---|---|
| `question` | Asks for information, evaluation, clarification, or a decision. |
| `delegation_request` | Assigns or requests concrete work from another agent. |
| `acknowledgement` | Confirms receipt or understanding without substantive work. |
| `status_update` | Reports progress, blockage, or current activity. |
| `result_delivery` | Delivers completed analysis, output, or requested work. |
| `evidence_observation` | Contributes facts, measurements, or observations. |
| `proposal` | Suggests a plan, hypothesis, design, or next action. |
| `critique_disagreement` | Challenges, rejects, or identifies a problem. |
| `agreement` | Explicitly accepts or endorses another contribution. |
| `decision` | Commits the group to a choice or conclusion. |
| `synthesis` | Combines multiple contributions into a unified account. |
| `clarification` | Resolves ambiguity or corrects interpretation. |
| `artifact_sharing` | Publishes or points to a file, link, run, or other artifact. |
| `coordination` | Manages sequencing, ownership, deadlines, or process. |
| `social_heartbeat` | Greeting, availability ping, or low-content presence update. |

**How types feed the three metrics:** primary and secondary types matching the scanner's relevant-type sets select a shared pool of candidate messages plus adjacent turns. The model then checks candidate text against all three failure definitions. A type is not itself a failure: for example, `result_delivery` does not establish that the result was integrated.

## 3. Collaboration metric definitions

These are **AgentTrace-defined, model-judged collaboration-failure categories**, not built-in Scout metrics or deterministic scores. The operational definitions below come from the scanner prompt.

| Category | Operational definition | Definition source |
|---|---|---|
| **Duplicated work**<br>`duplicated_work` | Different agents independently repeat materially equivalent analysis or execution without a clear verification, synthesis, or handoff purpose. **Exclude** acknowledgements, context summaries, and deliberate independent checks. | Scanner prompt lines 181-184 |
| **Weak handoff**<br>`weak_handoff` | Requested work is not acknowledged or delivered; delivered work is not integrated; ownership is unclear; or a request is abandoned. | Scanner prompt lines 185-186 |
| **Unintegrated information**<br>`unintegrated_information` | Relevant, visible information is later ignored, contradicted without resolution, or omitted from a decision despite a clear opportunity to use it. | Scanner prompt lines 187-189 |

Also summarized in MESSAGE_ANNOTATION.md: Collaboration findings. Source links target the main branch and may move as the implementation changes.

> **Interpretation matters.** `found=true` means the model judged the supplied evidence sufficient. `found=false` can mean insufficient evidence, not demonstrated absence. Confidence is model-reported, not a calibrated probability.
>
> Categories may overlap: nine traces produce 27 category judgments, not 27 distinct failures. Citation validation checks traceability, not semantic correctness. In the October 16 report, some saved rationales count purposeful verification as duplication despite the definition excluding it; human adjudication is needed.

# 4. Other analysis: distinct signals and labels

**Offline structural analysis** counts question marks and lexical agreement, challenge, and coordination markers, alongside lengths, timing, and turn structure. These are text-based signals, **not semantic message-type classifications**; they do not assign Scout's 15 types. See message-level metric definitions.

**Human audit codebook:** `commsfail` also provides the separate `discourse_v1` codebook below. These are human-labeling options, not an additional automatic Scout stage or results established by the report. Each label is a separate yes/no decision per post, unlike Scout's primary/secondary classification.

| Human discourse label | Codebook definition | Definition source |
|---|---|---|
| `accept` | The author makes an unconditional commitment to own a named item. Offers, intentions, and hedges do not count. | Codebook definition |
| `result` | The author asserts that a named item is delivered or ready for review. Label the assertion; whether the artifact exists is read from the record. | Codebook definition |
| `review_pass` | The author positively reviews another seat's current result for a named item. Intentions, questions, failures, and self-review do not count. | Codebook definition |

The separate regex_v1 annotator reports failure-mode evidence from text patterns; its failure-mode labels should not be confused with these message functions.

Prepared October 17, 2026. Self-contained HTML; no external scripts, fonts, or rendering dependencies.